



Build an App

05

Do you know that all the programs that you download and use in a smartphone are called applications? Let's learn more about them.



Over the years, computers and mobile phones have transformed from simple communication devices to functional tools. We are constantly on the lookout for products that make our lives more comfortable. Applications are one of these innovative products.

Mobile applications, or 'apps' as they are commonly called, have become an important feature in our day to day lives. An app exists for everything that we can think of, including finance, food, healthcare, navigation, etc. We can access these apps on smartphones anywhere with the help of mobile internet. Apps make smartphones fun and interesting and give us something to do when we're bored.



In the past, we had to follow a tedious process of going out to complete tasks like booking train tickets, catching a taxi, shopping, etc. Today, all these things can be done through an app.



App development follows a specific process:

- 1. Research** : Developers research the market with the help of surveys and studies to determine what kind of app the audience requires.
- 2. Wireframe** : A map or outline of the application's main idea is known as the wireframe.
- 3. Design** : Based on the application idea and the wireframe, the app's look and functions are designed.
- 4. Development** : After the look is created, developers work on the app's software, coding, and functionalities.
- 5. Distribution** : After it is completed, the app is distributed among the audience for use.
- 6. Maintenance** : The app is upgraded and maintained regularly to keep it relevant.



Let us look at the types of applications we commonly use:

- Utility apps help us organise our daily lives.

For example: Apps for contacts, alarms, reminders, to-do lists.



- Productivity apps make people more efficient while performing tasks. They also make some tasks easier.

For example: Calculators, Microsoft Office.



- Educational apps let us attend virtual classes from anywhere in the world in real time. They also provide recordings of previous classes. They help us submit assignments, watch tutorials, and improve our reading and listening skills.



- Banking apps let us access our bank and account information online. They help us send and receive money without having to go to a bank to make bill payments, transfers, or withdrawals.



- Social media apps are used to connect with people, share opinions, hobbies and interests online.



- Fitness apps allow users to set fitness goals, track food intake, gather workout ideas, and share their progress.



- News apps give us information from around the world in an instant, in the language that we choose. We don't have to wait for a newspaper anymore.

For example: Times of India app



- Entertainment apps like video streaming apps, eBooks, games, and music apps provide endless fresh entertainment through the internet.



Bytes

All mobile applications need a platform, known as an Operating System, where they are operated from. Android and iOS are the two most popular operating systems. All the above-mentioned types of applications can be created and operated on both, Android and iOS, separately.

5. Build an App

Apps can be used on other devices like laptops, tablets etc, that are easily accessible for people. Many different software can be used to create apps. One such software is MIT App Inventor.

What is MIT App Inventor?

- MIT App Inventor is an intuitive, visual programming environment that allows anyone to build fully functional apps for smartphones and tablets.
- It uses block-based tools to create apps.
- These tools can help create complex and engaging apps in significantly less time than traditional programming environments.
- MIT app inventor can be accessed online, via the website using a Gmail ID or offline, via software.



Let us show you how you can combine designing and coding to build an exquisite app.

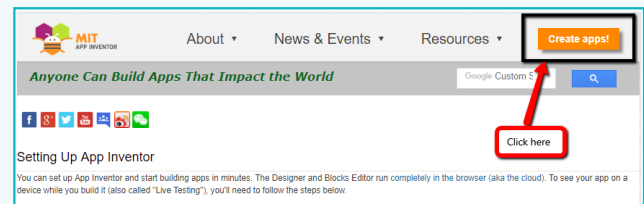


Activity

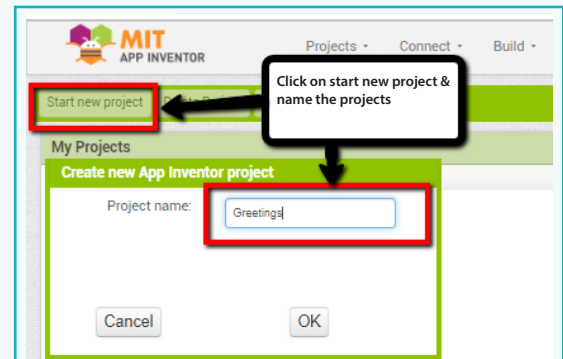
Use MIT app inventor to design and build a Doodle Cam application for Android OS.

1 Go to the MIT app inventor web page

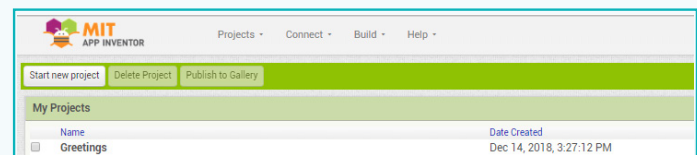
- To open to the MIT app inventor platform, open a browser window and enter the following link (internet required): <https://appinventor.mit.edu/>
- Go to 'Create apps!' on the right-hand top corner.



- Click "Continue" to dismiss the splash screen.
- To start the first project, click on "Start New Project". Add a project name and click "OK".



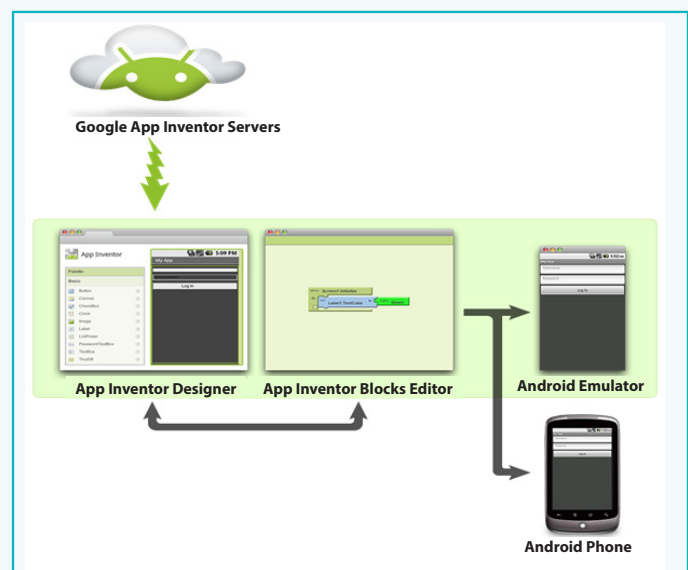
- The project will show up in the My Projects list. Click on it to open.



An app inventor consists of 3 programming components (windows):

- Designer
- Blocks Editor
- Android Emulator or Android Phone.

The double headed arrow in the image indicates that we can switch back and forth between the two sections while designing an app.

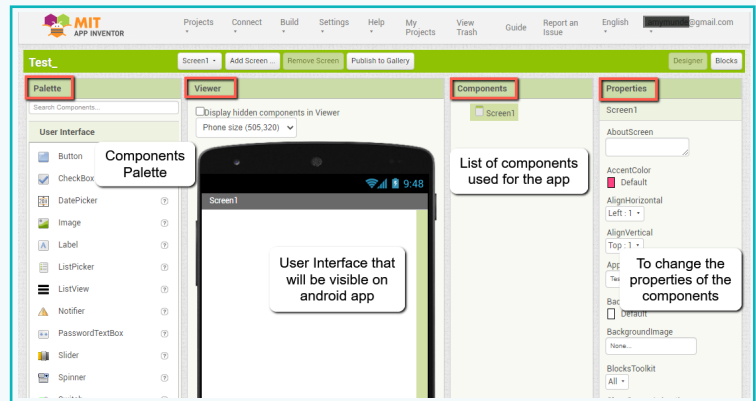


Activity

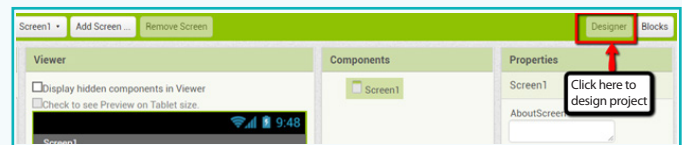
To build apps, we work with the app inventor designer, where we select **components** for the application & design the User Interface (UI).

- f. Look at the five sections of the design view labelled in the image (Palette, Viewer, Components, Media and Properties). The media section is visible below the components section.

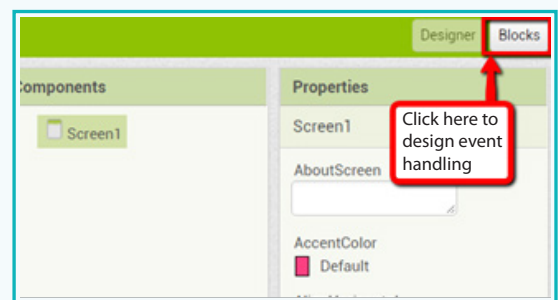
Note that, the media option in the components palette lets us add media-related components, while the media section in the design view lets us upload media files for use in the app.



- g. To start designing a project, place a component on the designer screen.

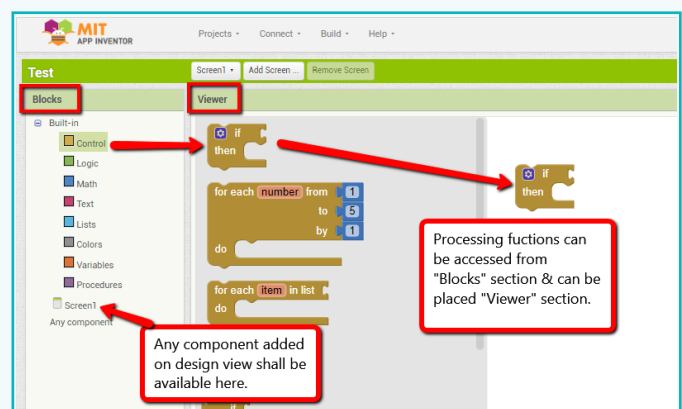


- h. To go to the coding part, click on the Blocks screen.



- i. To assemble program blocks, use the app inventor 'blocks' editor. Program blocks specify the functionality of the components used in the app.

There are two sections in the Blocks Editor (**Blocks and Viewer**) as shown below.



Activity

Below is a table that explains the terms that we just learned.

Sr. No.	Sub Parts (On project window)	Use
1	Designer	The Designer screen is where we create a visual representation of the app. It is also called User Interface (UI).
2	Blocks	In this, we design the functionality of the components we place in the Designer section.
3	Palette	The Palette has sub-palettes from where we place specified components into Designer or Blocks sections.
4	Viewer	A component appears in graphical manner in the Viewer.
5	Component	A Component is placed in the viewer screen. It makes up the parts of an app screen.
6	Properties	We can change component settings in Properties.

Exercise

State whether True or False:

- 1 MIT App Inventor can be accessed online, via website or offline, via software.
 _____
- 2 In the Designer section, we design the functionality of the components.
 _____
- 3 The media in the Palette section is different from the Media section of MIT inventor app.
 _____

Activity

- Create an app for marking up images taken with a smartphone camera using red, green, and blue colors, and save the edited images on the smartphone.

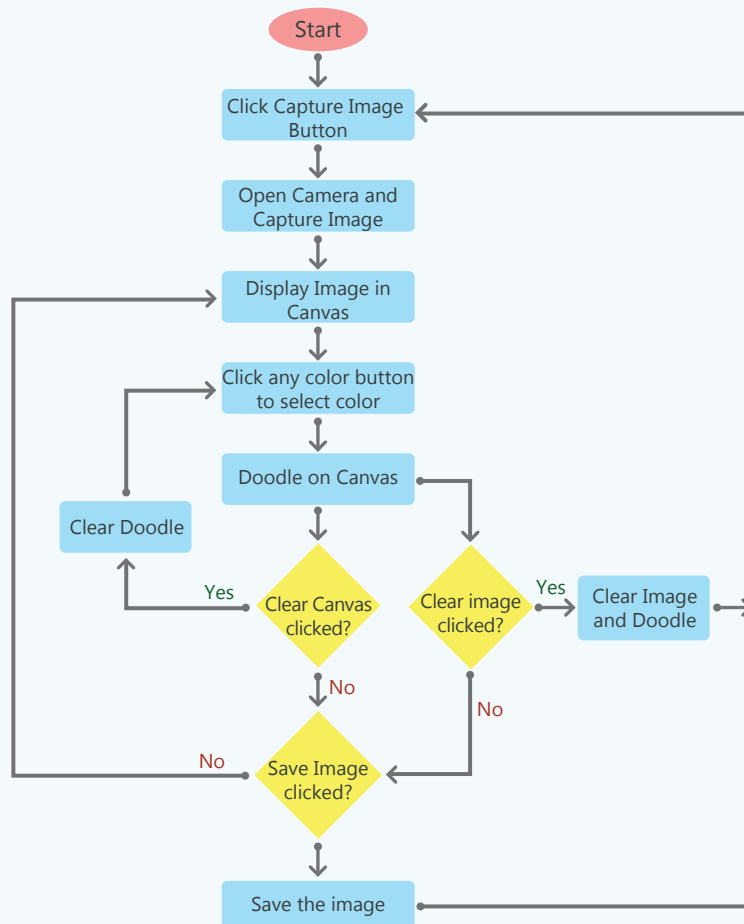
Steps to follow:

- Plan the flow of the app
- Identify required components for the UI
- Design the app's UI on MIT App Inventor
- Code its functionality

Flowchart:

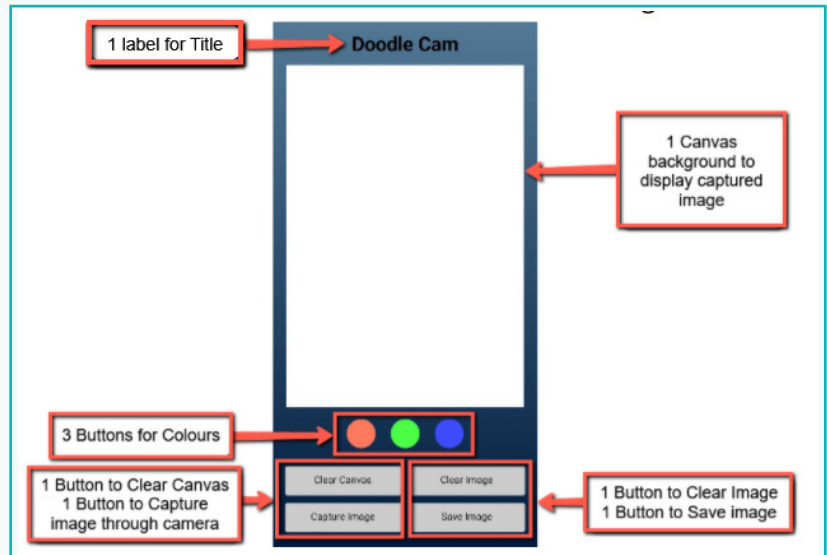
This flowchart shows us the process of capturing a photograph on a smartphone camera.

- To start, we click on the capture image button.
- We capture an image.
- After we click on the image, we can directly save the image or open it in canvas.
- We can doodle on the image in any colour.
- If we're happy with the doodles, we move ahead and save the image.
- If we do not like the doodles, we can clear them and try again.



Activity

UI Design: This is the canvas view where we can doodle on the image.

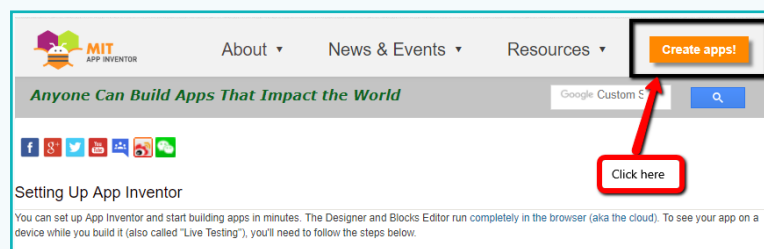


Designer Screen

1 Go to the web page:

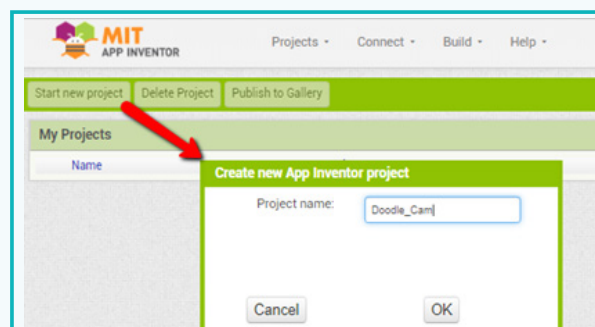
- <http://appinventor.mit.edu/explore/ai2/setup.html#>

Click “Create apps!” to start creating an application.



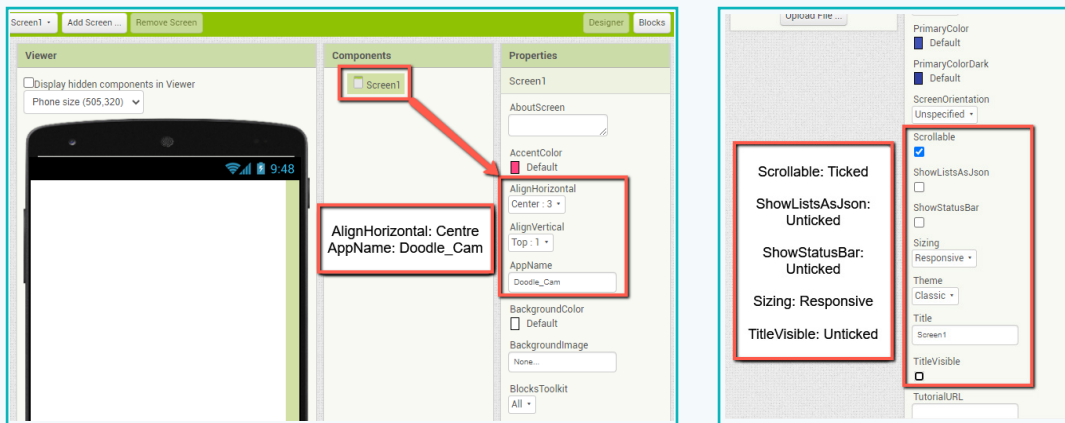
Click “Continue” to dismiss the splash screen.

2 To start a new project, click “Start new project”. Add a name for the project and click “OK”.



Activity

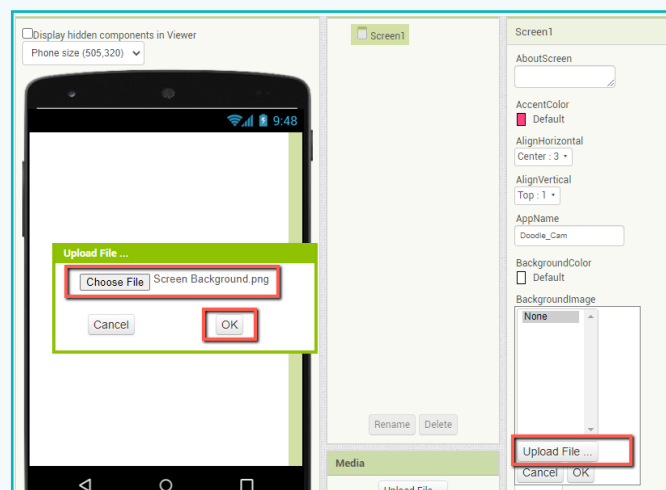
- Change Screen 1 properties as shown in the image. It will set the display of the app screens on any device where the app is installed.



- Set a background to highlight components on the screen.

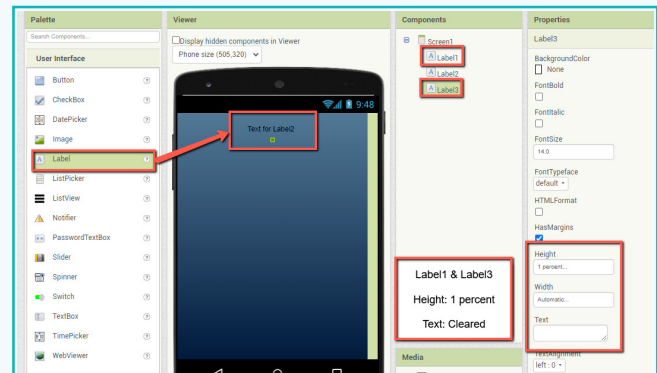
Note:

Search for a suitable background image from the internet to upload as a background for the screen. If the background image is unavailable, we can also set a background colour.

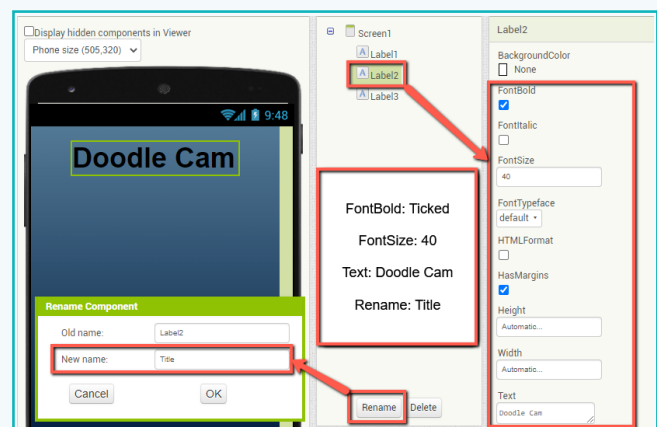


Activity

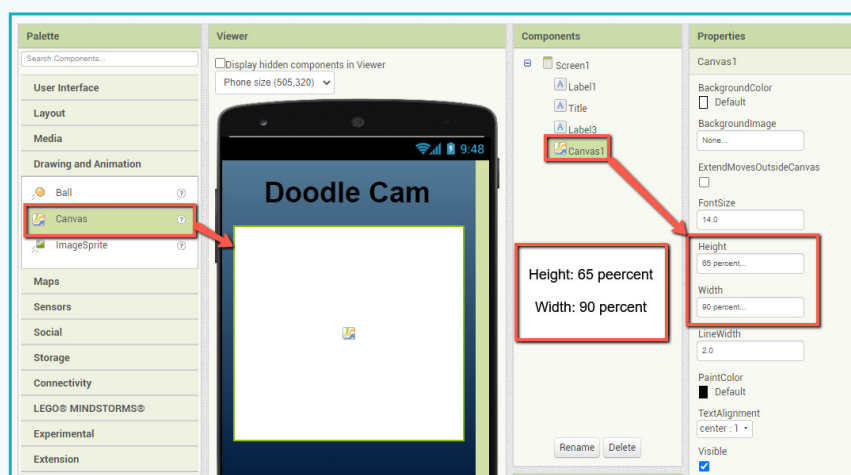
- Place 3 labels on the screen. Labels can be used as headings. They can also be used as spacing in between any two components. Here, Label1 & Label3 will be used as spacing labels.



- Set the properties of Label2 as shown and rename it “Title”.



- A container is required to place a captured picture or a blank image on the screen. Drag and drop a Canvas component from the ‘Drawing and Animation’ palette as shown. Change its height property to “65 percent” and weight property to “90 percent”. This will change the canvas size.



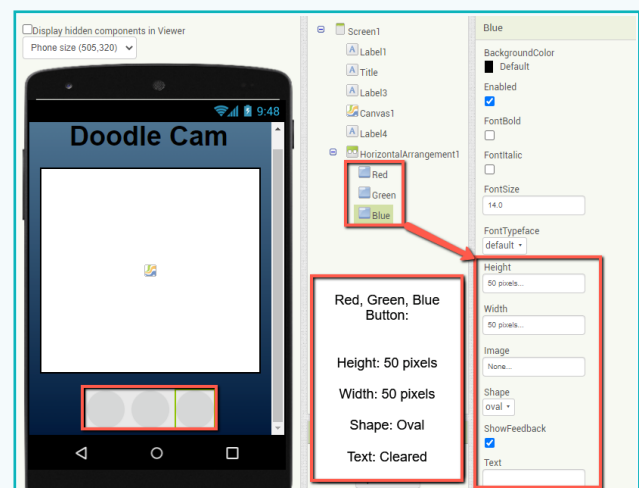
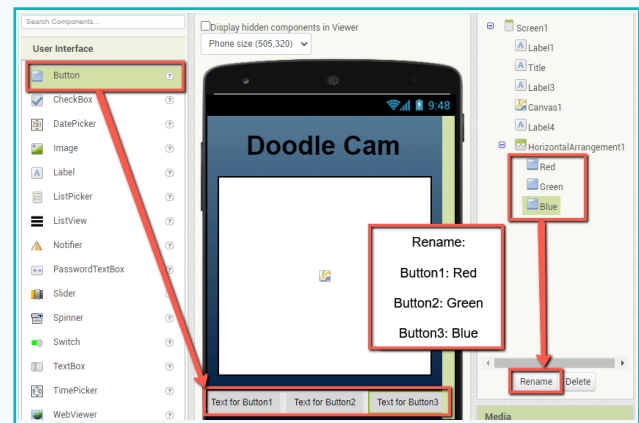
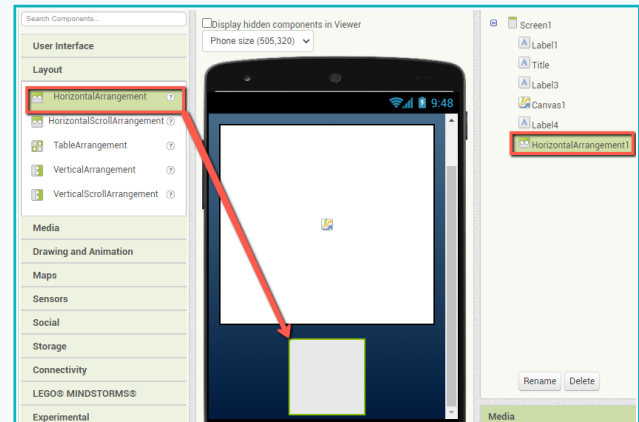
Activity

8 Place another label as a spacing label and change its properties as shown.

9 Place a “HorizontalArrangement” container from the Layout palette on Screen1 as shown. It will be used to horizontally align three colour buttons.

10 Drag and drop 3 buttons to “HorizontalArrangement”. Rename and change the button text to “Red”, “Green” and “Blue” respectively. These buttons will be used to select colours for doodling on the image.

11 Change the button properties as shown.

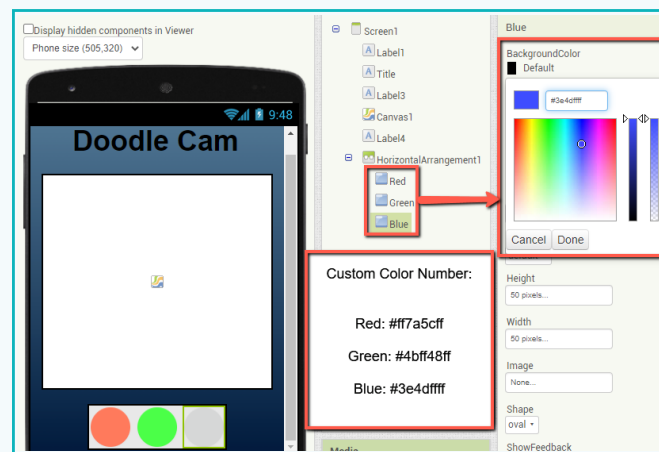


Activity

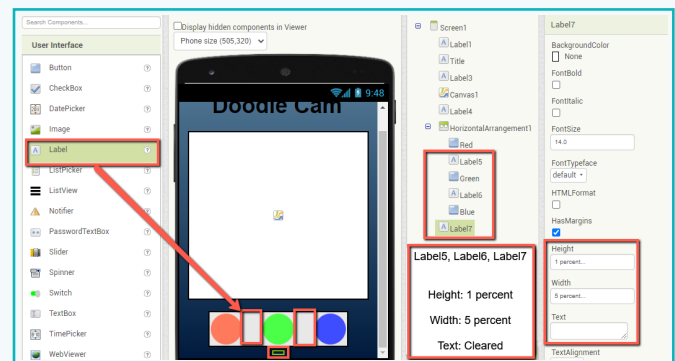
We can also change the background colours of the buttons from the properties section of each button.

12 Set custom background colours with the colour codes given below:

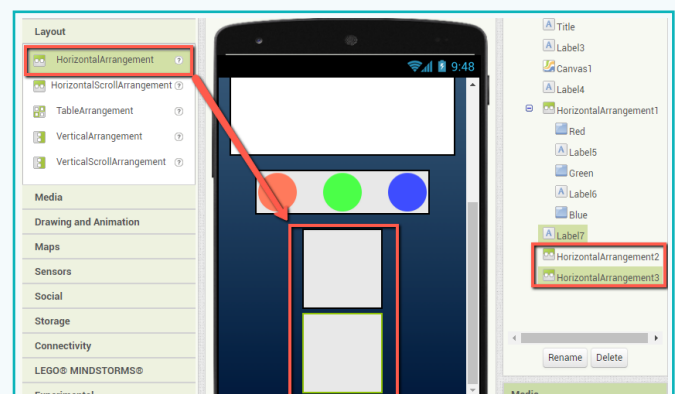
- Red: #ff7a5cff
- Green: #4bff48ff
- Blue: #3e4dffff



13 Place 2 labels for spacing in between the colour buttons. Place 1 label after HorizontalArrangement and change all their properties as shown.

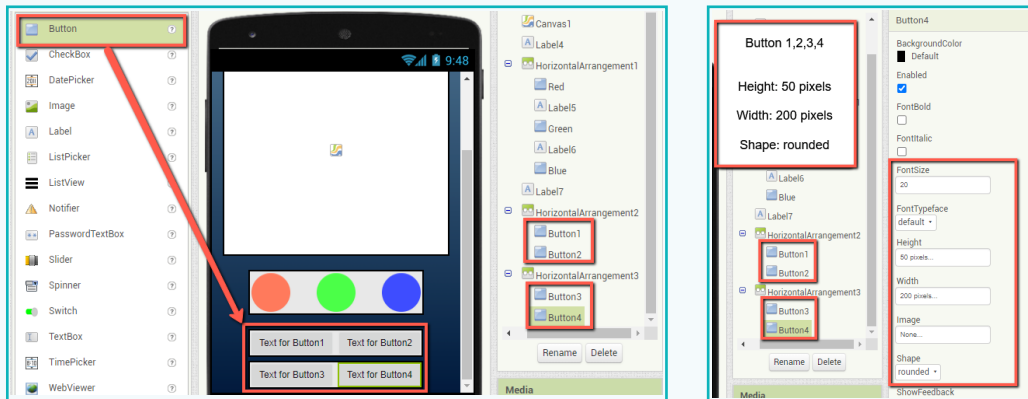


14 Place 2 “HorizontalArrangement” containers from the Layout palette on Screen1. These will be used to add 2 buttons in each container.

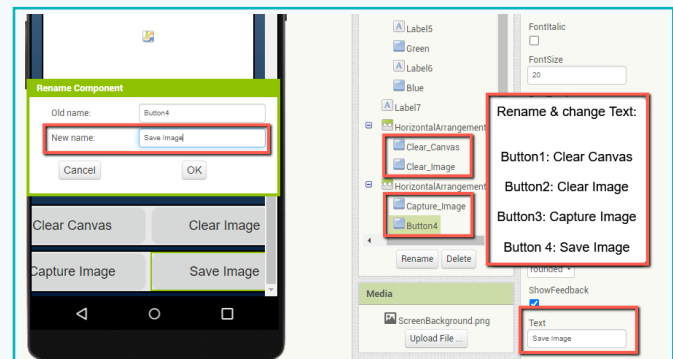


Activity

- 15 Place 2 buttons in “HorizontalArrangement2” and 2 buttons in “HorizontalArrangement3” and change their properties as shown.

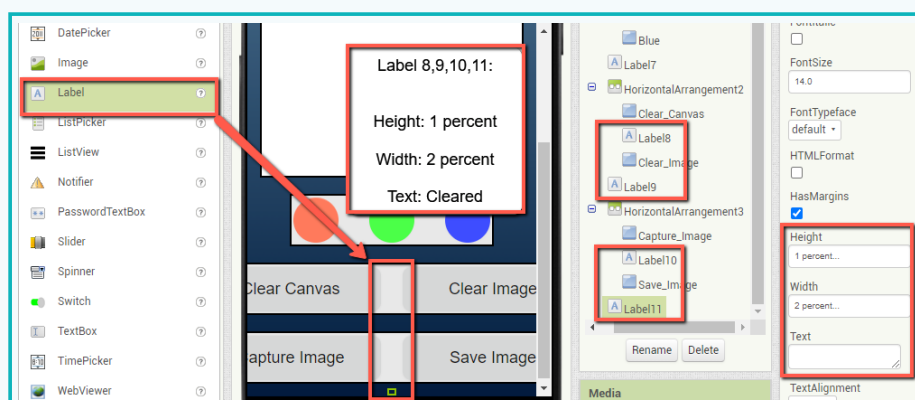


- 16 To identify the functionality of each button, rename it and change its text according to the image.



- 17 Place 1 label as a spacing label in between 2 buttons.

- 18 Place 1 label after each, HorizontalArrangement2 and HorizontalArrangement3, and change their properties as shown below.



Activity

Now that the UI of the app is ready, we need to build the functionality of the components placed in the app. That can be done from the Blocks screen.

Blocks Screen:

- 19 Click “Blocks” to switch to the Blocks Screen and program the functionality.

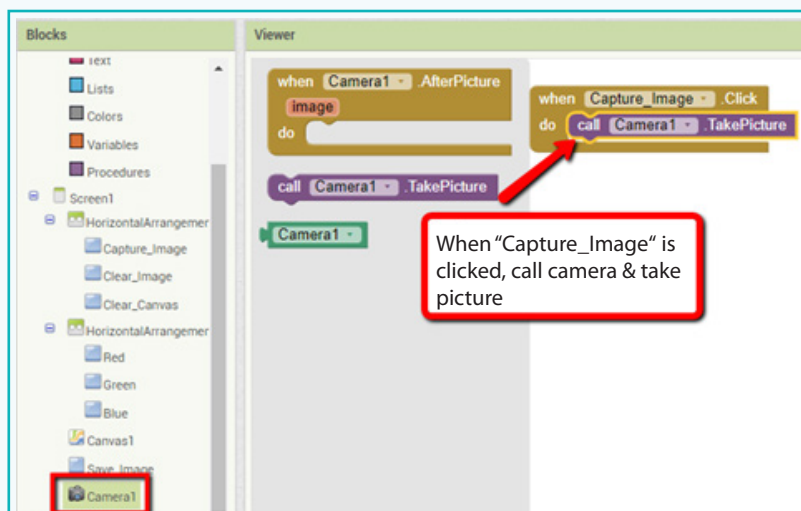


- 20 To ensure that a component is functional, click on the component to show all of its available events.

For example: To call the camera, place “when_Capture_Image.Click” from Capture_Image button.

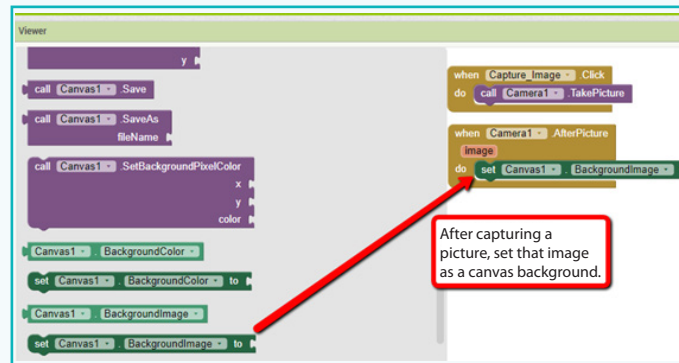


- 21 Connect “when_Capture_Image.Click” with “call.Camera.TakePicture” from the camera component.

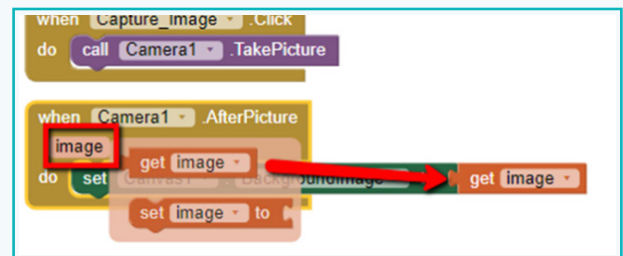


Activity

- 22 Add “set_Canvas1_BackgroundImage” block to “when_Camera1_AfterPicture”. Once the image is clicked, it will be set as canvas background.

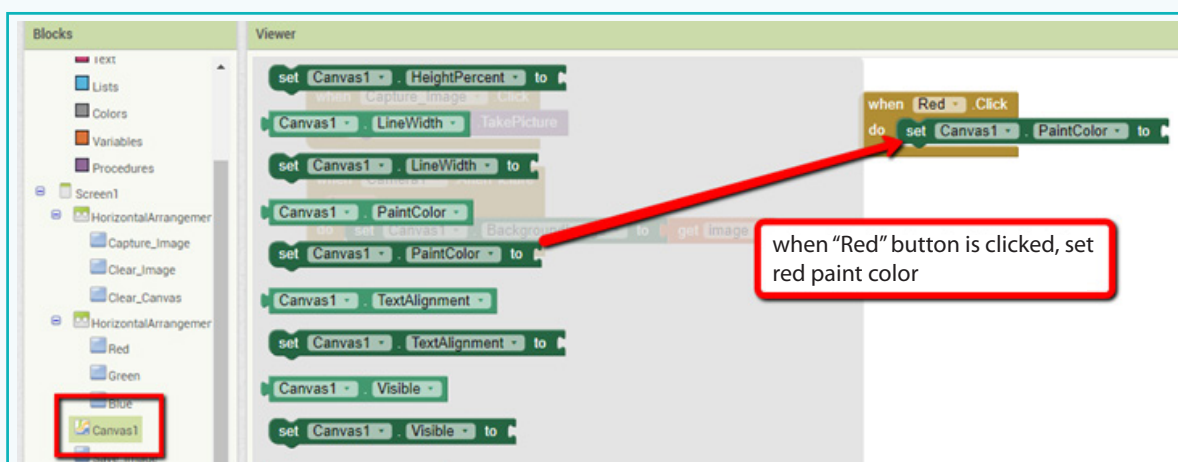


- 23 To display the captured Image, we have to connect “set_Canvas1_BackgroundImage” with the “get Image” block which is available on the “when_Camera1_AfterPicture” block. Follow the image shown.



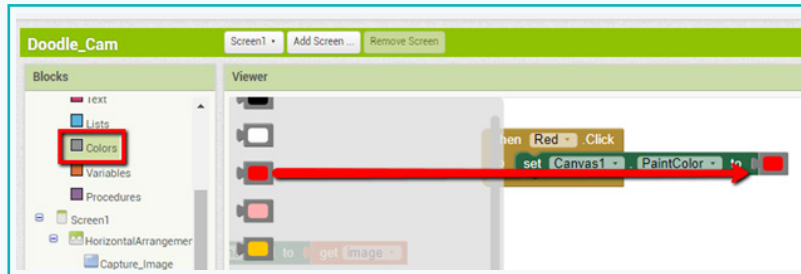
- 24 To code the functionality of colours in order to doodle, place “when_Red_Click” from the red component’s blocks.

- 25 Place “set_Canvas_PaintColor” from the canvas component under the red button event.

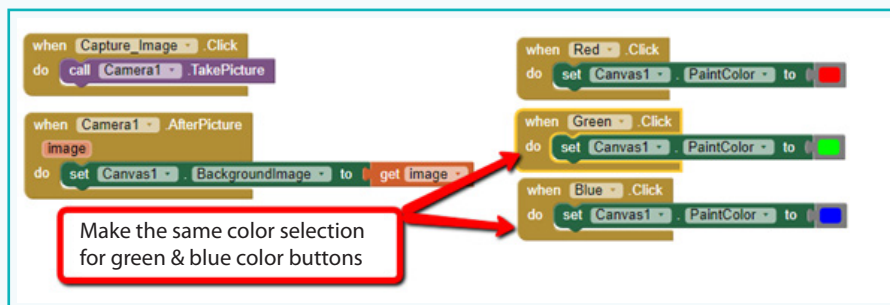


Activity

- 26 To set “Canvas_Paint color” to red, connect it with the red colour from the colour palette under the built-in blocks. It will set the colour of the canvas to red.

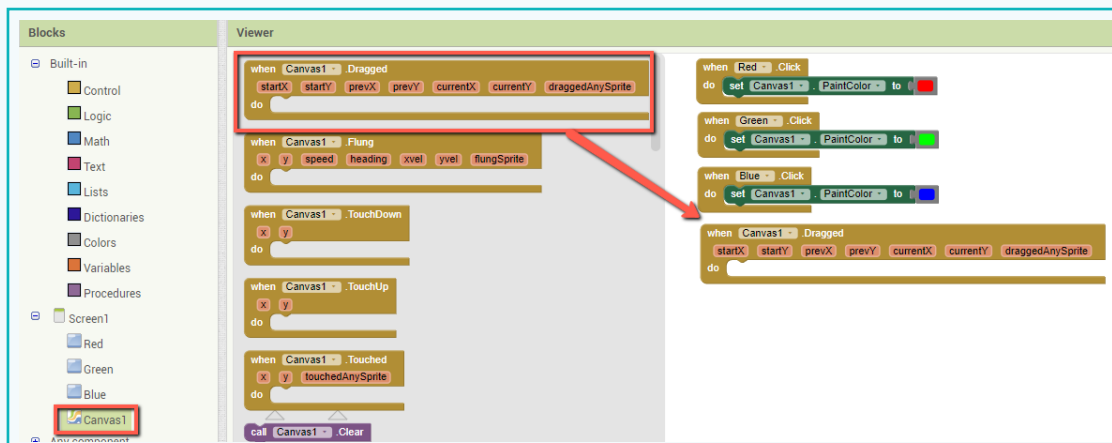


- 27 Make a similar block-based program for Green and Blue.

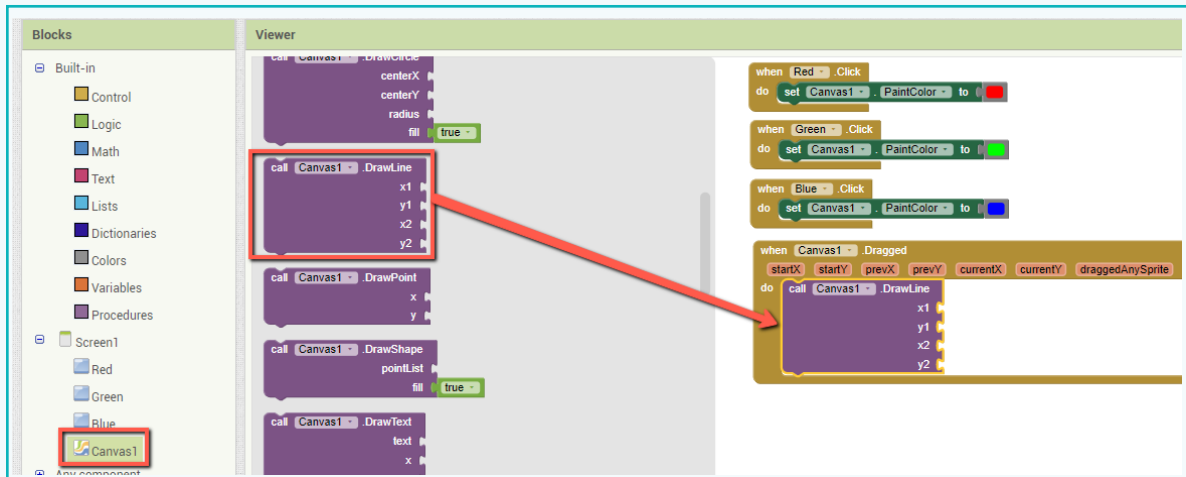


Line: A line is two distant points connected to each other. We implement the same logic to draw a line or a curve in the app. For this, every new point should be connected to its previous point. A point can be identified by X, Y coordinates.

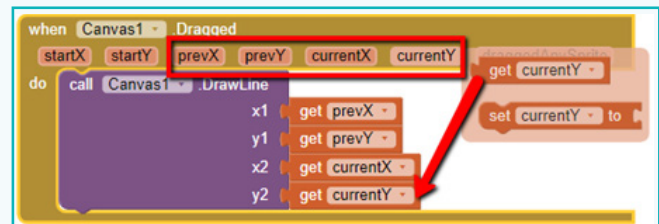
- 28 Place “when_Canvas1.Dragged” block of the canvas component and connect it with “call.Canvas1.DrawLine”. It will create a line from the start position (X1, Y1) to the end position (X2, Y2).



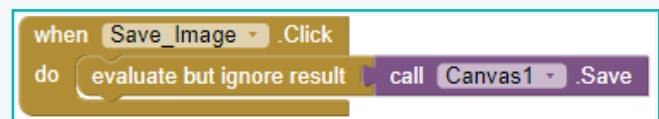
Activity



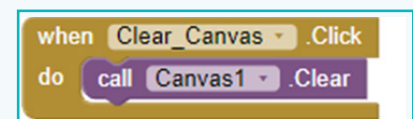
- 29 Draw a line block, connect “x1 to PrevX”, “y1 to PrevY”, “x2 to currentX” & “y2 to currentY”. It will create a line from the start position (X1, Y1) to the end position (X2, Y2).



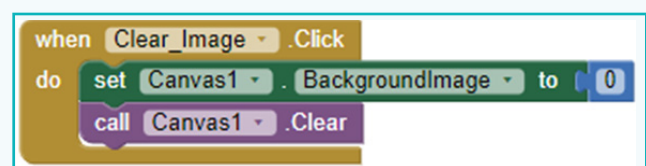
- 30 To save the image, connect “when_Save_Image.click” with “evaluate but ignore result” from the “Controls” blocks and “call.Canvas1.Save” from the Canvas component.



- 31 To clear the canvas, connect “when_Clear_Canvas.click” with “call.Canvas1.clear”.



- 32 To clear the image from the canvas, connect “when_Clear_Image.click” with “set.Canvas.BackgroundImage.to_0” and “call.Canvas1.Clear”.

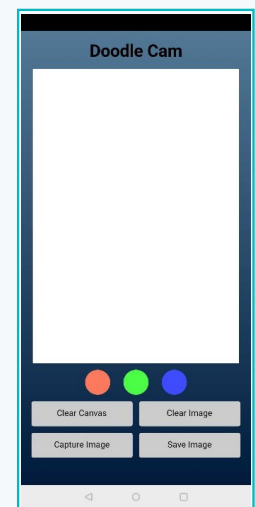
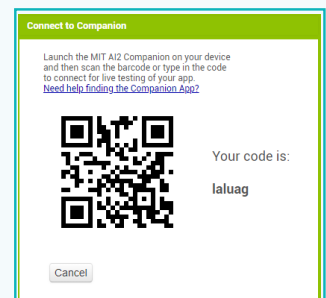
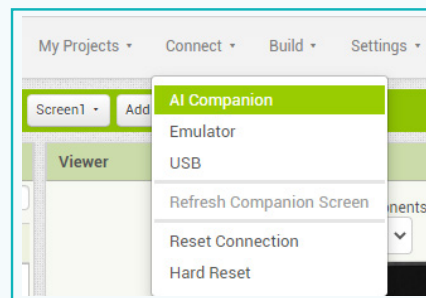
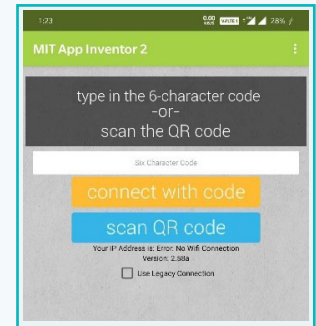


Now that we have finished building our app, we need to test if it works.

Activity

Test the functionality

- Open the MIT AI2 Companion app (should be pre-installed on the smartphone). This app will connect the smartphone with the app developed in the MIT app inventor so that we can test its functionality on the phone.
- Click “Scan QR code”.
- In the MIT app inventor, click Connect.
- Click AI Companion and scan the code.



The app will be uploaded to the smartphone where we can test its functionality.

Apps have become a part and parcel of our daily lives. They have made life easy and made entertainment more accessible. With the help of app inventor programs, we can successfully create our own apps. This can help solve a problem faced by different people. In this lesson, we used the MIT app inventor to create a Doodle Cam application.

The app we have created will help us take visual notes and mark important aspects in an image which can be saved for future reference. It also allows us to make creative doodles for fun.

Yes! And we can also make fun doodles with it.

We can use this app to conveniently take notes and mark an image for reference.



Tech Flash

- **MIT App Inventor** : An intuitive, visual programming environment that allows anyone to build fully functional apps for smartphones and tablets.
- **Component Designer** : The platform where components of the app can be used to design the User Interface (UI) of the app.
- **Blocks Editor** : The platform where program blocks can be assembled to specify the functionality of the components.